

VARANASI SATHVEEK

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Education

National Institute of Technology Andhra Pradesh

Aug. 2023 – May 2027

Bachelor of Technology in Computer Science and Engineering – CGPA: 8.76/10.0

Tadepalligudem, Andhra Pradesh

Relevant Coursework: Applied Machine Learning, Natural Language Processing, Probability & Statistics, Data Structures & Algorithms, Design & Analysis of Algorithms, Database Management Systems, Operating Systems, Computer Networks

Experience

Intants Pvt Ltd

May 2026 – Present

Software Engineer Intern

Remote

- Automated extraction, cleaning, validation, and transformation of vendor invoice data on a full-stack platform (Python/FastAPI, PostgreSQL, React), eliminating manual data-entry effort across the finance workflow.
- Designed a normalized PostgreSQL schema and wrote optimized SQL queries for invoice ingestion and line-item reconciliation, improving data accuracy and processing throughput under concurrent load.

National Institute of Technology Andhra Pradesh

Aug 2025 – Dec 2025

Data & Backend Developer Intern

Tadepalligudem, Andhra Pradesh

- Built data analytics pipelines in Python (pandas, NumPy) to process user activity and performance data, and delivered interactive dashboards visualizing topic-level trends, reducing student preparation time by ~30%.
- Designed a question prioritization and tagging system driven by analysis of user profiles and historical performance, powering personalized recommendations that improved practice efficiency by ~20%.

Indian Institute of Technology Ropar

May 2025 – Aug 2025

Research Intern

Ropar, Punjab

- Modeled large relational datasets as graphs to analyze relationships and dependencies, surfacing network patterns and structural insights that were not visible in raw tabular data.
- Built an interactive visualization interface with filtering and toggling for real-time exploration of relational data, improving insight discovery and usability for researchers.

Projects

MediLens – Biomedical Question Answering | PyTorch, HuggingFace, FAISS, FastAPI, Docker



- Built a retrieval-augmented pipeline over PubMed: downloaded, parsed, and chunked abstracts into a 262K-chunk corpus, embedded them with a PubMedBERT sentence encoder, and indexed them in FAISS for dense retrieval.
- Fine-tuned BioBERT and PubMedBERT for biomedical named-entity recognition, and chained retrieval, cross-encoder reranking, and generation (BioGPT/Flan-T5, local models via Ollama) into a single pipeline returning structured JSON.
- Evaluated answers with ROUGE-L and embedding similarity against published BioGPT, PubMedGPT, and ClinicalBERT scores; deployed a Dockerized FastAPI service, substituting a TF-IDF retriever to fit a 512MB instance.

Mechanistic Interpretability of Transformers | PyTorch, TransformerLens, GPT-2



- Ran a series of interpretability experiments on GPT-2 Small: activation caching, attention pattern analysis, residual stream decomposition, and induction-head detection.
- Used activation patching as a causal intervention to identify which attention heads influence specific predictions, building a working understanding of transformer internals; manuscript in preparation.

Technical Skills

Languages: Python, SQL, C++, Java, JavaScript, TypeScript, Bash

ML & Deep Learning: PyTorch, HuggingFace Transformers, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, TransformerLens; transformers, sequence models

LLMs & Retrieval: Prompting, RAG, fine-tuning, FAISS, sentence-transformers, cross-encoder reranking, local models via Ollama, Gemini API

Databases & Tools: PostgreSQL, MySQL, MongoDB, SQLite, Git, Docker, AWS, REST APIs, FastAPI, Jupyter, React.js, Node.js

Achievements / Leadership

- Contest rating of **1570** on LeetCode; ranked in the **Top 1.6%** of JEE Mains (17,801 / 1.1M candidates).
- Led Techkriya sponsorships, securing INR 1 lakh through industry outreach and stakeholder presentations; awarded the Meritorious Idea Award by the Institute Innovation Council.